ARTIFICIAL INTELLIGENCE (CT 653)

Tribhuvan University | Institute of Engineering | BCT | Year III / Part II

CHAPTER 4: Knowledge Representation, Inference and Reasoning

Complete Exam-Ready Answers | All Past Questions Covered

CHAPTER 4 SYLLABUS OVERVIEW [8 hrs]

According to the syllabus, Chapter 4 covers:

- 4.1 Formal logic – connectives, truth tables, syntax, semantics, tautology, validity, well-formed-formula
- 4.2 Propositional logic, predicate logic, FOPL, interpretation, quantification, horn clauses
- 4.3 Rules of inference, unification, resolution refutation system (RRS), answer extraction from RRS, rule-based deduction system
- 4.4 Statistical Reasoning – Probability and Bayes' theorem and causal networks, reasoning in belief network

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SECTION 1: Propositional Logic & Predicate Logic (FOPL)

Syllabus Reference: 4.1, 4.2 | Exam Ref: 2079 Ashwin Back, 2070 Magh, 2071 Magh, 2077 Chaitra

Q1. What is knowledge and learning? Why are they important? Explain the factors required to pass the Turing Test. [2+2+4]

Answer:

Knowledge: In AI, knowledge refers to the information, facts, rules, and relationships that an intelligent system possesses to make decisions and solve problems. It is the foundation of reasoning and intelligent behavior.

Learning: Learning is the process by which an agent improves its performance through experience. It allows a system to acquire new knowledge, refine existing knowledge, and adapt to new situations without being explicitly programmed for each case.

Importance:

- Knowledge enables an agent to understand the environment and make informed decisions.
- Learning allows the agent to improve over time and handle situations not anticipated by the programmer.
- Together, they are essential for building truly intelligent systems that can operate in complex, real-world environments.

Factors Required to Pass the Turing Test: The Turing Test, proposed by Alan Turing (1950), tests whether a machine can exhibit intelligent behavior indistinguishable from a human. To pass it, a machine must have:

1. Natural Language Processing (NLP) – ability to understand and generate human language.
2. Knowledge Representation – ability to store information about the world.
3. Automated Reasoning – ability to use stored information to answer questions and draw conclusions.
4. Machine Learning – ability to adapt to new situations and detect patterns.
5. Computer Vision – ability to perceive images and objects (for Total Turing Test).
6. Robotics – ability to interact physically with the world (for Total Turing Test).

Q2. What is Propositional Logic? What is Predicate Logic (FOPL)? Explain with examples, including syntax, semantics, tautology, validity, and well-formed formula. [4+4]

Answer: Propositional Logic

Propositional Logic (also called Propositional Calculus or Boolean Logic) deals with propositions — statements that are either TRUE or FALSE. It uses logical connectives to combine propositions.

Connectives and Truth Table:

Connective	Symbol	Meaning	Example
Negation	¬	NOT	¬P = 'It is not raining'
Conjunction	∧	AND	P ∧ Q = 'It rains and road is wet'
Disjunction	∨	OR	P ∨ Q = 'It rains or it is cloudy'
Implication	→	IF...THEN	P → Q = 'If it rains, road is wet'
Biconditional	↔	IF AND ONLY IF	P ↔ Q = 'It rains iff road is wet'

Well-Formed Formula (WFF): A syntactically correct formula in propositional logic. Rules:

- Every propositional symbol (P, Q, R...) is a WFF.
- If P is a WFF, then ¬P is a WFF.
- If P and Q are WFFs, then (P ∧ Q), (P ∨ Q), (P → Q), (P ↔ Q) are all WFFs.
- Nothing else is a WFF.

Tautology: A formula that is TRUE under ALL possible interpretations (truth assignments). Example: P ∨ ¬P is always TRUE.

Validity: A formula is valid if it is a tautology — true in every model. Example: (P → Q) ∧ P → Q (Modus Ponens) is valid.

Satisfiability: A formula is satisfiable if it is TRUE under at least one interpretation. P ∧ Q is satisfiable (when both P and Q are true).

Answer: First Order Predicate Logic (FOPL)

FOPL (also called First-Order Logic or Predicate Calculus) extends propositional logic by introducing predicates, variables, constants, functions, and quantifiers. It can express relationships between objects.

Components of FOPL:

Component	Description	Example
Predicate	A function that returns TRUE or FALSE	Human(Socrates), Loves(x,y)
Constant	A specific named object	Socrates, Nepal, 5
Variable	A placeholder for objects	x, y, z
Function	Maps objects to objects	father(x), sqrt(4)
Quantifier ∀	Universal – 'for all'	∀x Human(x) → Mortal(x)
Quantifier □	Existential – 'there exists'	□x Loves(x, Mary)

Example Sentences in FOPL:

- 'All humans are mortal': ∀x Human(x) → Mortal(x)
- 'Socrates is a human': Human(Socrates)
- 'Socrates is mortal': Mortal(Socrates) [derived by Modus Ponens]
- 'Someone loves Mary': □x Loves(x, Mary)
- 'No dog is a cat': ∀x Dog(x) → ¬Cat(x)

Horn Clauses: A Horn clause is a clause with at most ONE positive literal. It takes the form: P1 ∧ P2 ∧ ... ∧ Pn → Q. Horn clauses are the basis of Prolog and efficient reasoning systems.

SECTION 2: Rules of Inference & Unification

Syllabus Reference: 4.3 | Exam Ref: 2076 Baisakh, 2078 Chaitra, 2079 Ashwin, 2080 Ashwin, 2075 Baisakh

Q3. List down all the Rules of Inference. When can we use these rules? Explain each with suitable example. [3+5]

Answer:

Rules of Inference are logical rules that allow us to derive new facts (conclusions) from existing known facts (premises) in a valid and sound way.

Rule	Form	Description	Example
Modus Ponens	$P, P \to Q \vdash Q$	If P is true and P→Q, then Q is true	Rains→Wet. Rains. ∴ Wet.
Modus Tollens	$\neg Q, P \to Q \vdash \neg P$	If Q is false and P→Q, then P is false	¬Wet, Rains→Wet. ∴ ¬Rains.
Hypothetical Syllogism	$P \to Q, Q \to R \vdash P \to R$	Chain implications	P→Q, Q→R ∴ P→R
Disjunctive Syllogism	$P \vee Q, \neg P \vdash Q$	If one is false, other is true	Rain∨Snow, ¬Rain ∴ Snow
And-Introduction	$P, Q \vdash P \wedge Q$	Combine two true statements	'Ram smart', 'Ram tall' ∴ 'Ram smart and tall'
And-Elimination	$P \wedge Q \vdash P$	Extract one part of AND	'Ram smart and tall' ∴ 'Ram smart'
Or-Introduction	$P \vdash P \vee Q$	Add any disjunct	'It rains' ∴ 'It rains or snows'
Double Negation	$\neg\neg P \vdash P$	Remove double negation	¬¬Rains ⊢ Rains
Universal Instantiation	$\forall x\, P(x) \vdash P(a)$	Apply universal fact to specific	∀x Mortal(x). ∴ Mortal(Socrates)
Existential Generalization	$P(a) \vdash \exists x\, P(x)$	Generalize from specific	Tall(Ram) ∴ ∃x Tall(x)
Resolution	$P \vee Q, \neg Q \vee R \vdash P \vee R$	Core rule for theorem proving	(P∨Q) and (¬Q∨R) → (P∨R)

When to use these rules: Rules of inference are used whenever we need to derive new knowledge from existing knowledge in logical reasoning, automated theorem proving, expert systems, and FOPL-based AI systems.

Q4. What is Unification? Explain the Unification Algorithm with an example. [3+4]

Answer:

Unification: Unification is the process of finding a substitution (called the Most General Unifier - MGU) that makes two or more logical expressions identical. It is a key component of resolution-based theorem proving and Prolog.

Unification Algorithm (Robinson's Algorithm):

1. Start with the pair of expressions to unify. Set substitution θ = {} (empty).
2. If both expressions are identical, return θ (success).
3. If one is a variable x and the other is a term t, and x does not occur in t, add {x/t} to θ and apply it.
4. If both are compound expressions with same predicate/function and arity, unify each pair of arguments.
5. If none of the above applies, FAIL (expressions cannot be unified).

Examples:

- Unify P(x) and P(John): θ = {x/John} → Result: P(John) ✓
- Unify P(x, y) and P(John, Mary): θ = {x/John, y/Mary} → P(John, Mary) ✓
- Unify P(x, f(x)) and P(John, f(John)): θ = {x/John} → P(John, f(John)) ✓
- Unify P(x, x) and P(John, Mary): FAIL — cannot set x to both John and Mary
- Unify P(x) and P(f(x)): FAIL — 'Occurs Check' fails (x appears in f(x))

SECTION 3: Resolution Refutation System (RRS) — Most Repeated Topic

Syllabus Reference: 4.3 | Exam Ref: 2082 Kartik, 2081 Ashwin, 2080 Chaitra, 2079 Ashwin Back, 2078 Poush, 2077 Chaitra, 2075 Baisakh, 2075 Bhadra, 2073 Bhadra, 2072 Magh, etc.

Q5. What is CNF (Conjunctive Normal Form)? Why is CNF necessary? Explain ALL steps to convert a FOPL statement to CNF with a suitable example. [Repeated in nearly every set] [2+6]

Answer:

Conjunctive Normal Form (CNF): A formula is in CNF if it is expressed as a CONJUNCTION (AND) of CLAUSES, where each clause is a DISJUNCTION (OR) of literals (a literal is an atomic formula or its negation).

General Form: $(L11 \vee L12 \vee ...) \wedge (L21 \vee L22 \vee ...) \wedge ...$

Example of CNF: $(A \vee B) \wedge (\neg C \vee D) \wedge (E \vee \neg F \vee G)$

Why CNF is Necessary: The Resolution rule can only be applied to clauses in CNF. Converting to CNF allows us to use the Resolution Refutation Method to prove theorems automatically in FOPL.

Steps to Convert to CNF:

Step 1 — Eliminate Biconditional (↔): Replace $P \leftrightarrow Q$ with $(P \rightarrow Q) \wedge (Q \rightarrow P)$

Step 2 — Eliminate Implication (→): Replace $P \rightarrow Q$ with $\neg P \vee Q$

Step 3 — Move Negation Inward using De Morgan's Laws:

- $\neg(P \wedge Q) = \neg P \vee \neg Q$
- $\neg(P \vee Q) = \neg P \wedge \neg Q$
- $\neg(\forall x)\, P(x) = (\exists x)\, \neg P(x)$
- $\neg(\exists x)\, P(x) = (\forall x)\, \neg P(x)$
- $\neg\neg P = P$

Step 4 — Standardize Variables: Rename variables so each quantifier uses a unique variable name. No two quantifiers should share the same variable.

Step 5 — Skolemize (Remove Existential Quantifiers):

- Replace each existential variable with a Skolem constant (if no universal quantifier precedes it) or a Skolem function.
- Rule: If $\exists x$, no preceding $\forall$ → replace x with constant: P(a)
- Rule: If $\forall y\, \exists x\, P(x,y)$ → replace x with Skolem function of y: P(f(y), y)

Step 6 — Drop Universal Quantifiers: Remove all $\forall$ quantifiers (all remaining variables are assumed universally quantified).

Step 7 — Distribute OR (∨) over AND (∧): Apply: $A \vee (B \wedge C) = (A \vee B) \wedge (A \vee C)$ to get CNF.

Full Example: 'Everyone who loves all animals is loved by someone'

Original English: Everyone who loves all animals is loved by someone.

FOPL: $\forall x\, [\forall y\, Animal(y) \rightarrow Loves(x,y)] \rightarrow [\exists z\, Loves(z,x)]$

Step 1 (No biconditionals): Already done.

Step 2 (Eliminate →): $\forall x\, [\neg\forall y\, (\neg Animal(y) \vee Loves(x,y))] \vee [\exists z\, Loves(z,x)]$

Step 3 (Move ¬ inward): $\forall x\, [\exists y\, (Animal(y) \wedge \neg Loves(x,y))] \vee [\exists z\, Loves(z,x)]$

Step 4 (Standardize): $\forall x\, [\exists y\, (Animal(y) \wedge \neg Loves(x,y))] \vee [\exists z\, Loves(z,x)]$ (y and z are already distinct)

Step 5 (Skolemize): y → F(x) [Skolem function], z → G(x) [Skolem function]

```
[Animal(F(x)) ∧ ¬Loves(x,F(x))] ∨ Loves(G(x),x)
```

Step 6 (Drop ∀x): [Animal(F(x)) ∧ ¬Loves(x,F(x))] ∨ Loves(G(x),x)

Step 7 (Distribute ∨ over ∧):

- Clause 1: Animal(F(x)) ∨ Loves(G(x),x)
- Clause 2: ¬Loves(x,F(x)) ∨ Loves(G(x),x)

These two clauses are the CNF representation.

Q6. Use Resolution Refutation to solve: The law says that it is a crime for an American to sell weapons to hostile nations. The country Nono, an enemy of America, has some missiles, and all of its missiles were sold to it by Colonel West, who is American. Prove That Colonel West is criminal. [Most Repeated — 2082, 2081, 2077, 2075, 2073, 2071, etc.] [8]

Answer:

Step 1 — FOPL Representation of Facts:

1. It is a crime for an American to sell weapons to hostile nations:
   ```
   ∀x ∀y ∀z [American(x) ∧ Weapon(y) ∧ Sells(x,y,z) ∧ Hostile(z)] → Criminal(x)
   ```
2. Nono has missiles:
   ```
   ∃x Owns(Nono, x) ∧ Missile(x)
   ```
3. All of Nono's missiles were sold by West:
   ```
   ∀x Missile(x) ∧ Owns(Nono,x) → Sells(West, x, Nono)
   ```
4. Missiles are weapons:
   ```
   ∀x Missile(x) → Weapon(x)
   ```
5. An enemy of America is hostile:
   ```
   ∀x Enemy(x, America) → Hostile(x)
   ```
6. West is American:
   ```
   American(West)
   ```
7. Nono is an enemy of America:
   ```
   Enemy(Nono, America)
   ```

Negated Goal (what we want to disprove):

```
¬Criminal(West)
```

Step 2 — Convert to CNF Clauses:

- C1: ¬American(x) ∨ ¬Weapon(y) ∨ ¬Sells(x,y,z) ∨ ¬Hostile(z) ∨ Criminal(x)
- C2: Owns(Nono, M1) [Skolem constant M1 for missile]
- C3: Missile(M1)
- C4: ¬Missile(x) ∨ ¬Owns(Nono,x) ∨ Sells(West, x, Nono)
- C5: ¬Missile(x) ∨ Weapon(x)
- C6: ¬Enemy(x, America) ∨ Hostile(x)
- C7: American(West)
- C8: Enemy(Nono, America)
- C9: ¬Criminal(West) [Negated goal]

Step 3 — Resolution Proof:

Step	Clauses Resolved	Substitution	Result

1	C3 + C5	{x/M1}	Weapon(M1)
2	C6 + C8	{x/Nono}	Hostile(Nono)
3	C3 + C2 + C4	{x/M1}	Sells(West, M1, Nono)
4	C1 + C7	{x/West}	¬Weapon(y) ∨ ¬Sells(West,y,z) ∨ ¬Hostile(z) ∨ Criminal(West)
5	Step4 + Step3	{y/M1, z/Nono}	¬Weapon(M1) ∨ ¬Hostile(Nono) ∨ Criminal(West)
6	Step5 + Step1	{}	¬Hostile(Nono) ∨ Criminal(West)
7	Step6 + Step2	{}	Criminal(West)
8	Step7 + C9	{}	□ (Empty Clause — CONTRADICTION)

Conclusion: Since we derived a contradiction (empty clause □), the negated goal ¬Criminal(West) is FALSE. Therefore, COLONEL WEST IS A CRIMINAL is PROVED. ✓

Q7. Assume the following facts: Every child loves Santa. Everyone who loves Santa loves any reindeer. Rudolph is a reindeer and has a red nose. Anything with a red nose is weird or is a clown. No reindeer is a clown. Scrooge does not love anything which is weird. Prove: Scrooge is not a child. [2081 Chaitra, 2078 Chaitra] [8]

Answer:

Step 1 — FOPL Representation:

1. Every child loves Santa: ∀x Child(x) → Loves(x, Santa)
2. Everyone who loves Santa loves any reindeer: ∀x ∀y Loves(x,Santa) ∧ Reindeer(y) → Loves(x,y)
3. Rudolph is a reindeer: Reindeer(Rudolph)
4. Rudolph has a red nose: RedNose(Rudolph)
5. Anything with a red nose is weird or a clown: ∀x RedNose(x) → Weird(x) ∨ Clown(x)
6. No reindeer is a clown: ∀x Reindeer(x) → ¬Clown(x)
7. Scrooge does not love anything weird: ∀x Weird(x) → ¬Loves(Scrooge, x)

Negated Goal: ¬(¬Child(Scrooge)) = Child(Scrooge) [assume Scrooge IS a child and derive contradiction]

Step 2 — CNF Clauses:

- C1: ¬Child(x) ∨ Loves(x, Santa)
- C2: ¬Loves(x,Santa) ∨ ¬Reindeer(y) ∨ Loves(x,y)
- C3: Reindeer(Rudolph)
- C4: RedNose(Rudolph)
- C5: ¬RedNose(x) ∨ Weird(x) ∨ Clown(x)
- C6: ¬Reindeer(x) ∨ ¬Clown(x)
- C7: ¬Weird(x) ∨ ¬Loves(Scrooge, x)
- C8: Child(Scrooge) [negated goal]

Step 3 — Resolution:

Step	Resolved	Result

1	C4 + C5 {x/Rudolph}	Weird(Rudolph) ∨ Clown(Rudolph)
2	C3 + C6 {x/Rudolph}	¬Clown(Rudolph)
3	Step1 + Step2	Weird(Rudolph)
4	C8 + C1 {x/Scrooge}	Loves(Scrooge, Santa)
5	Step4 + C2 {x/Scrooge, y/Rudolph}	¬Reindeer(Rudolph) ∨ Loves(Scrooge, Rudolph)
6	Step5 + C3	Loves(Scrooge, Rudolph)
7	Step3 + C7 {x/Rudolph}	¬Loves(Scrooge, Rudolph)
8	Step6 + Step7	□ (Contradiction)

Conclusion: Contradiction reached. Our assumption that Scrooge is a child is FALSE. Therefore, SCROOGE IS NOT A CHILD is PROVED. ✓

Q8. Assume: Horses, cows, pigs are mammals. An offspring of a horse is a horse. Bluebeard is a horse. Bluebeard is Charlie's parent. Offspring and parent are inverse relations. Every mammal has a parent. Prove Charlie is a horse using Resolution Refutation. [2079 Jestha, 2078 Poush, 2078 Baishakh, 2073 Bhadra] [8]

Answer:

FOPL Statements:

1. Horse(Bluebeard) ∧ Cow(Bluebeard) ∧ Pig(Bluebeard) — well, Horse(x)∨Cow(x)∨Pig(x) → Mammal(x)
2. ∀x ∀y Horse(x) ∧ Offspring(y,x) → Horse(y)
3. Horse(Bluebeard)
4. Parent(Bluebeard, Charlie)
5. ∀x ∀y Parent(x,y) ↔ Offspring(y,x)
6. ∀x Mammal(x) → ∃ y Parent(y,x)

Negated Goal: ¬Horse(Charlie)

CNF Clauses:

- C1: ¬Horse(x) ∨ ¬Offspring(y,x) ∨ Horse(y)
- C2: Horse(Bluebeard)
- C3: Parent(Bluebeard, Charlie)
- C4: ¬Parent(x,y) ∨ Offspring(y,x) [from biconditional]
- C5: ¬Horse(Charlie) [negated goal]

Resolution:

Step	Clauses	Result
1	C3 + C4 {x/Bluebeard, y/Charlie}	Offspring(Charlie, Bluebeard)
2	C2 + C1 {x/Bluebeard}	¬Offspring(y,Bluebeard) ∨ Horse(y)
3	Step1 + Step2 {y/Charlie}	Horse(Charlie)
4	Step3 + C5	□ (Contradiction)

Conclusion: Contradiction derived. Therefore, CHARLIE IS A HORSE is PROVED. ✓

SECTION 4: Forward Chaining, Backward Chaining & Production Systems

Syllabus Reference: 4.3 | Exam Ref: 2082 Kartik, 2081 Ashwin, 2080 Chaitra, 2078 Poush, 2077 Chaitra, 2072 Ashwin, 2070 Magh

Q9. Define Production System. Define and differentiate Forward Chaining and Backward Chaining. [2082 Kartik, 2081 Ashwin, 2077 Chaitra, 2072 Ashwin] [2+4+4]

Answer: Production System

A Production System is a model of computation in AI that consists of three components:

1. Working Memory (Database): Contains the current facts/state of the world.
2. Production Rules (Rule Base): A set of condition-action rules of the form: IF <condition> THEN <action>.
3. Inference Engine: Controls which rules to apply and in what order (conflict resolution).

Example Rule: IF temperature > 100 AND patient has rash THEN patient may have fever (diagnosis).

Forward Chaining (Data-Driven / Bottom-Up Reasoning)

Forward Chaining starts from KNOWN FACTS and applies inference rules to derive NEW FACTS, continuing until the GOAL is reached or no more rules apply.

Algorithm:

1. Place all known facts in Working Memory.
2. Check if any rule's condition (LHS) is satisfied by current facts.
3. If yes, fire the rule — add its conclusion (RHS) to Working Memory.
4. Repeat until goal is reached or no new facts can be derived.

Example:

- Facts: American(West), Missile(M1), Owns(Nono,M1), Enemy(Nono,America)
- Rule: Missile(x) $\wedge$ Owns(Nono,x) $\rightarrow$ Sells(West,x,Nono)
- Fire $\rightarrow$ Add: Sells(West,M1,Nono)
- Rule: Enemy(x,America) $\rightarrow$ Hostile(x). Fire $\rightarrow$ Hostile(Nono)
- Rule: American(x)$\wedge$Weapon(y)$\wedge$Sells(x,y,z)$\wedge$Hostile(z) $\rightarrow$ Criminal(x)
- Fire $\rightarrow$ Conclude: Criminal(West) ✓ GOAL REACHED

Backward Chaining (Goal-Driven / Top-Down Reasoning)

Backward Chaining starts from the GOAL and works BACKWARD, breaking it into sub-goals, until it finds facts that are already known to be true.

Algorithm:

1. Set the goal as the initial query.
2. Find rules whose conclusion (RHS) matches the goal.
3. The conditions (LHS) of that rule become new sub-goals.
4. Recursively prove each sub-goal (check database or apply more rules).
5. If all sub-goals are satisfied, the original goal is proved.

Example: Goal: Criminal(West)?

- Need: American(West) $\wedge$ Weapon(y) $\wedge$ Sells(West,y,Nono) $\wedge$ Hostile(Nono)
- American(West)? $\rightarrow$ Found in DB ✓
- Weapon(y)? $\rightarrow$ Need Missile(y) $\rightarrow$ Found: Missile(M1) $\rightarrow$ Weapon(M1) ✓
- Sells(West,M1,Nono)? $\rightarrow$ Need Missile(M1)$\wedge$Owns(Nono,M1) $\rightarrow$ Both found ✓
- Hostile(Nono)? $\rightarrow$ Need Enemy(Nono,America) $\rightarrow$ Found ✓

- All sub-goals satisfied → Criminal(West) PROVED ✓

Comparison Table:

Feature	Forward Chaining	Backward Chaining
Direction	Facts → Goal (Bottom-Up)	Goal → Facts (Top-Down)
Starting Point	Known facts in working memory	The goal/hypothesis
Driven By	Data / Evidence	Goal / Query
Used When	Goal is not known in advance	Specific goal is known
May Derive	All possible conclusions (even irrelevant)	Only relevant conclusions
Efficiency	May be slow (derives unnecessary facts)	More focused and efficient
Used In	Planning, monitoring, CLIPS	Diagnosis, Prolog, expert systems
Example	CLIPS rule engine	Prolog, medical diagnosis

SECTION 4: Statistical Reasoning, Bayes' Theorem & Belief Networks

Syllabus Reference: 4.4 | Exam Ref: 2081 Ashwin, 2080 Ashwin, 2079 Ashwin Back, 2078 Chaitra, 2078 Baishakh, 2075 Bhadra, 2073 Magh, 2071 Magh

Q10. What are the approaches for measuring uncertainty? 1% of people have a certain genetic defect. 90% of tests for the gene detect the defect (true positives). 10% of the tests are false positives. a) What is the probability of getting a positive test result? b) If a person gets a positive test result, what are the odds they actually don't have the genetic defect? [2081 Ashwin] [2+6]

Answer: Approaches for Measuring Uncertainty

In AI, uncertainty arises because we may not have complete or perfect information. Key approaches include:

1. Probability Theory: Assigns numerical probabilities to uncertain events ($0 \le P \le 1$).
2. Bayesian Reasoning: Updates probability given new evidence using Bayes' Theorem.
3. Fuzzy Logic: Deals with degrees of truth (partial truth values between 0 and 1).
4. Dempster-Shafer Theory: Assigns belief and plausibility measures to uncertain information.
5. Rule-Based Certainty Factors: Used in expert systems (e.g., MYCIN) to represent partial confidence.

Solution using Bayes' Theorem:

Bayes' Theorem Formula:

```
P(H|E) = [ P(E|H) × P(H) ] / P(E)
```

Given:

- P(Defect) = 0.01 (Prior: 1% of people have the defect)
- P(No Defect) = 0.99
- P(Positive | Defect) = 0.90 (True positive rate)
- P(Positive | No Defect) = 0.10 (False positive rate)

Part a) P(Positive test result): Using the Total Probability Theorem:

```
P(Positive) = P(Pos|Defect) × P(Defect) + P(Pos|No Defect) × P(No Defect)
            = (0.90 × 0.01) + (0.10 × 0.99)
            = 0.009 + 0.099
            = 0.108
```

Answer: The probability of getting a positive test = 0.108 = 10.8%

Part b) P(No Defect | Positive):

```
P(No Defect | Pos) = P(Pos | No Defect) × P(No Defect) / P(Positive)
                   = (0.10 × 0.99) / 0.108
                   = 0.099 / 0.108
                   ≈ 0.917 = 91.7%
```

Answer: Even with a positive test result, there is a ~91.7% chance the person does NOT have the genetic defect. This demonstrates the importance of prior probability in Bayesian reasoning.

Q11. What is Prior Probability and Posterior Probability? Show how these are used in Bayes Theorem. Is there any connection between Belief Network and Bayes Theorem? Justify with an example. [2080 Chaitra] [3+5]

Answer:

Prior Probability P(H): The probability of a hypothesis H BEFORE observing any evidence. It represents our initial belief based on background knowledge. Example: P(Cancer) = 0.001 — 0.1% of population has cancer.

Posterior Probability P(H|E): The probability of a hypothesis H AFTER observing evidence E. It is the updated belief. Example: P(Cancer | Positive Test) = ? — after seeing a positive test result.

Likelihood P(E|H): The probability of observing evidence E given that hypothesis H is true. Example: P(Positive Test | Cancer) = 0.95

Bayes' Theorem:

```
P(H|E) = [ P(E|H) × P(H) ] / P(E)
```

This formula allows us to update our belief (prior) to a new belief (posterior) after seeing evidence.

Belief Network (Bayesian Network): A Belief Network (also called Bayesian Network or Causal Network) is a directed acyclic graph (DAG) where:

- Each NODE represents a random variable.
- Each EDGE represents a direct causal relationship / dependency.
- Each node has a Conditional Probability Table (CPT) giving P(node | parents).

Connection to Bayes' Theorem: Belief Networks encode Bayes' Theorem across multiple variables. They allow efficient computation of joint probabilities using the chain rule and conditional independence:

```
P(X1, X2, ..., Xn) = ∏ P(Xi | Parents(Xi))
```

Classic Example — 'Wet Grass' Bayesian Network [2078 Baishakh]:

Nodes: Cloudy (C), Sprinkler (S), Rain (R), Wet Grass (W)

- C → S, C → R, S → W, R → W
- P(C) = 0.6 (given)
- P(S|C=T) = 0.10, P(S|C=F) = 0.50
- P(R|C=T) = 0.80, P(R|C=F) = 0.20
- P(W|S=T,R=T) = 0.99, P(W|S=T,R=F) = 0.90
- P(W|S=F,R=T) = 0.90, P(W|S=F,R=F) = 0.01

Query: Find P(W=T | C=T, S=T, R=F)

```
P(W=T | S=T, R=F) = 0.90   [from CPT directly]
```

Answer: P(Wet Grass = True | Cloudy=T, Sprinkler=T, Rain=F) = 0.90

Q12. How reasoning is done in uncertainty? Three factories produce light bulbs. Factory A produces 20%, Factory B produces 50%, Factory C produces 30%. 2% of A's bulbs are defective, 5% of B's, 3% of C's. A bulb is selected at random and found defective. What is the probability it was produced by Factory B? [2080 Ashwin Back] **[3+5]**

Answer:

Given:

- P(A) = 0.20, P(B) = 0.50, P(C) = 0.30
- P(D|A) = 0.02, P(D|B) = 0.05, P(D|C) = 0.03

Find: P(B|D) — probability bulb is from B given it is defective.

Step 1 — Total Probability of Defective P(D):

```
P(D) = P(D|A)×P(A) + P(D|B)×P(B) + P(D|C)×P(C)
     = (0.02×0.20) + (0.05×0.50) + (0.03×0.30)
     = 0.004 + 0.025 + 0.009
```

```
        = 0.038
```

Step 2 — Apply Bayes' Theorem:

```
P(B|D) = P(D|B) × P(B) / P(D)
       = (0.05 × 0.50) / 0.038
       = 0.025 / 0.038
       ≈ 0.6579 = 65.79%
```

Conclusion: The probability that the defective bulb was produced by Factory B is approximately 65.79%.

SECTION 5: Semantic Networks, Frames & Knowledge Representation

Syllabus Reference: 4.2, 5.1–5.5 (closely related) | Exam Ref: 2082, 2081, 2079, 2078, 2077, 2076, 2075, 2073, 2072, 2071

Q13. Define term Frame. Convert the following sentences into Semantic Network. Cat is a mammal. Cat has fur. Mammal has vertebra. Mammal is an animal. Bear is a mammal. Bear has fur. Whale is a mammal. Whale lives in water. Fish is an animal. Fish lives in water. [2082 Kartik, 2078 Poush] [1+6]

Answer: Frame

A FRAME is a structured data representation for knowledge. It is a record-like structure that contains:

- A SLOT for each attribute of the object.
- A FILLER (value) for each slot.
- Default values that can be overridden.
- Inheritance through a class hierarchy.

Example Frame for 'Cat':

Frame: CAT	
ISA	Mammal
Has-part	Fur
Sound	Meow
Legs	4
Habitat	Land

Semantic Network for the Given Sentences:

A Semantic Network is a graphical representation of knowledge as nodes (concepts/objects) connected by labeled edges (relationships).

Relationships used: ISA (is a/is a kind of), HAS (has property), LIVES-IN

The Semantic Network has the following nodes and links:

From Node	Relationship	To Node
Cat	ISA	Mammal
Cat	HAS	Fur
Mammal	HAS	Vertebra
Mammal	ISA	Animal
Bear	ISA	Mammal
Bear	HAS	Fur
Whale	ISA	Mammal
Whale	LIVES-IN	Water
Fish	ISA	Animal

Fish	LIVES-IN	Water

Textual representation of the Network:

```
            ANIMAL
           /      \
         ISA      ISA
         /          \
     MAMMAL         FISH
    / | |  \        |
  ISA |  |  ISA  LIVES-IN
   /  |  |   \     |
  Cat Bear Whale  Water
  |    |    |
  HAS HAS LIVES-IN
  |    |    |
  Fur Fur  Water
```

Note: Bear inherits Vertebra from Mammal through ISA link (inheritance property of semantic networks).

Q14. What are the different approaches for knowledge representation? How do you make a choice among these approaches? List down any issues associated with it. [2080 Chaitra] [2+2+2+2]

Answer: Approaches to Knowledge Representation

Approach	Description	Example
Logical Representation	Uses formal logic (propositional/predicate) to represent facts and rules	∀x Human(x)→Mortal(x)
Semantic Networks	Graph-based: nodes are concepts, edges are relationships	Dog ISA Animal, Dog HAS Legs
Frames	Record-like structures with slots and fillers	Frame: Car [Color: Red, Speed: 100]
Production Rules	IF-THEN rules in a rule base	IF fever AND rash THEN measles
Scripts	Sequence of events for stereotyped situations	Restaurant Script: Enter, Sit, Order...
Conceptual Dependency	Representation of actions and their semantic meaning	John ATRANS book to Mary

How to Choose an Approach:

- If the domain has clear logical rules → use Logical Representation (FOPL).
- If the domain involves categorization and inheritance → use Semantic Networks or Frames.
- If the domain requires event-based reasoning → use Scripts.
- If reasoning is rule-driven → use Production Rules.

Issues in Knowledge Representation:

1. Breadth of Coverage: Difficult to represent ALL knowledge about the world.
2. Ambiguity: Natural language is ambiguous — hard to represent precisely.
3. Efficiency: Large knowledge bases are slow to search.
4. Incompleteness: No system can represent all knowledge; some facts may be missing.
5. Inconsistency: Conflicting rules or facts can arise.
6. Knowledge Acquisition Bottleneck: Getting experts to encode knowledge is slow and expensive.

QUICK REVISION: Chapter 4 — Most Important Points Summary

Topic	Key Points	Marks Weight
Propositional Logic	Connectives (∧,∨,¬,→,↔), Truth Table, WFF, Tautology, Validity	4-6
FOPL	Predicates, Constants, Variables, Functions, ∀, □, Horn Clauses	4-6
CNF Conversion	7 Steps: Eliminate ↔ → Move ¬ in → Standardize → Skolemize → Drop ∀ → Distribute	6-8
Resolution Refutation	Negate goal → CNF → Resolve until □ (empty clause)	8
Rules of Inference	Modus Ponens, Tollens, Syllogism, Universal Instantiation, Resolution	3-5
Unification	Find MGU to make two expressions identical	4-6
Forward Chaining	Data→Goal, data-driven, may derive irrelevant facts	4-6
Backward Chaining	Goal→Data, goal-driven, focused, used in Prolog	4-6
Bayes' Theorem	P(H|E) = P(E|H)×P(H)/P(E) — update belief with evidence	6-8
Belief Networks	DAG with CPTs, encodes conditional independence, uses Bayes	4-6
Semantic Networks	Graph: nodes=concepts, edges=relations (ISA, HAS, LIVES-IN)	4-6
Frames	Record structure with slots, fillers, defaults, inheritance	2-4

Best of Luck in Your Examination!

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